

HDOC DAY-12 REPORT

Menu-Driven Digit Operations Using Switch-Case

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Activity	HDOC Day-12

Prepared according to the Day-12 instructions: algorithm, test-case table, C program, compilation, execution and testing.

Question / Problem Statement

Write a menu-driven C program using **switch-case** to input a positive integer and perform one of the following operations based on the user's choice:

1. Find the sum and product of the first and last digits. Example: Number = 12345, Sum = 6, Product = 5.
2. Count how many digits are less than, equal to, and greater than a given digit **k**. Example: Number = 12345, k = 3, Count_less = 2, Count_equal = 1, Count_greater = 2.

Algorithm

Step 1: Start.

Step 2: Display two menu choices.

Step 3: Read the user's choice.

Step 4: Use switch-case.

Step 5: If choice is 1, input a positive integer. Get the last digit using $n \% 10$. Repeatedly divide the number by 10 until the first digit remains. Calculate and display the sum and product of the first and last digits. **Step**

6: If choice is 2, input a positive integer and digit k. Extract each digit using $n \% 10$, compare it with k, and increment the less/equal/greater counter. Divide by 10 and repeat until all digits are processed.

Step 7: For any other choice, display 'Invalid choice'.

Step 8: Stop.

Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Choice = 1 Number = 12345	Sum = 6 Product = 5	Sum = 6 Product = 5	Pass
2	Choice = 2 Number = 12345 k = 3	Count_less = 2 Count_equal = 1 Count_greater = 2	Count_less = 2 Count_equal = 1 Count_greater = 2	Pass
3	Choice = 5	Invalid choice	Invalid choice	Pass

Operation 1 - Program Screenshot

```
OMYADAV@OMs-WINDOWS 11 ~ % clang day12.1.c -o day12.1
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1
Enter a positive integer: 1
Sum = 2
Product = 1
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1
Enter a positive integer: 3
Sum = 6
Product = 9
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1c
zsh: no such file or directory: ./day12.1c
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1
Enter a positive integer: 3
Sum = 6
Product = 9
OMYADAV@OMs-WINDOWS 11 ~ %
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1
Enter a positive integer: 3
Sum = 6
Product = 9
OMYADAV@OMs-WINDOWS 11 ~ % █
```

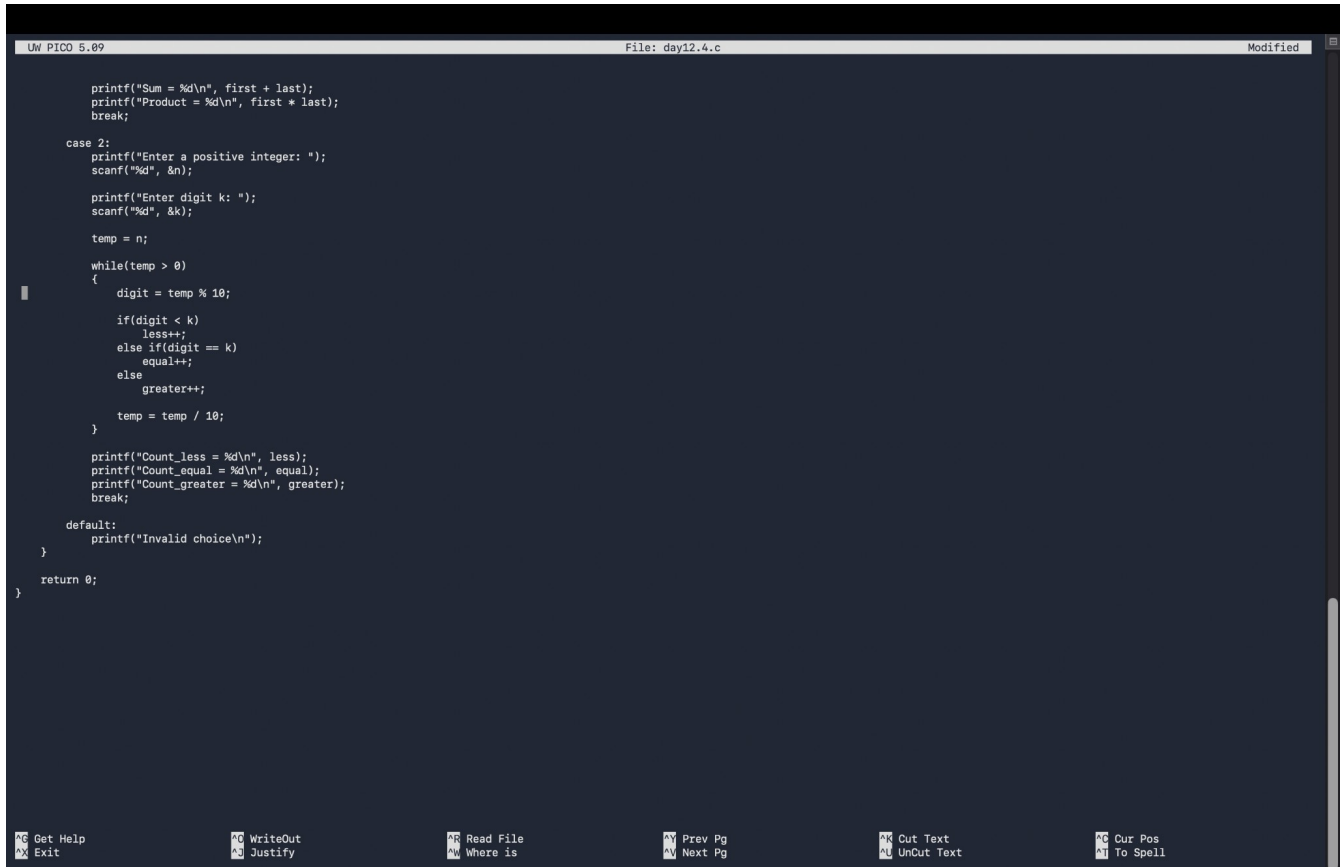
C program used to find the sum and product of the first and last digits.

Operation 1 - Compilation and Execution

```
OMYADAV@OMs-WINDOWS 11 ~ % clang day12.1.c -o day12.1
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1
Enter a positive integer: 1
Sum = 2
Product = 1
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1
Enter a positive integer: 3
Sum = 6
Product = 9
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1c
zsh: no such file or directory: ./day12 1c
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1
Enter a positive integer: 3
Sum = 6
Product = 9
OMYADAV@OMs-WINDOWS 11 ~ % █
```

Execution evidence for the first operation. The program was compiled and tested with positive integers.

Operation 2 - Compilation and Execution



The screenshot shows a code editor window titled "UW PICO 5.09" with the file "day12.4.c" open. The code is a C program that calculates the sum and product of two numbers, then counts the number of digits less than, equal to, and greater than a given digit k. The code is as follows:

```
printf("Sum = %d\n", first + last);
printf("Product = %d\n", first * last);
break;

case 2:
    printf("Enter a positive integer: ");
    scanf("%d", &n);

    printf("Enter digit k: ");
    scanf("%d", &k);

    temp = n;

    while(temp > 0)
    {
        digit = temp % 10;

        if(digit < k)
            less++;
        else if(digit == k)
            equal++;
        else
            greater++;

        temp = temp / 10;
    }

    printf("Count_less = %d\n", less);
    printf("Count_equal = %d\n", equal);
    printf("Count_greater = %d\n", greater);
    break;

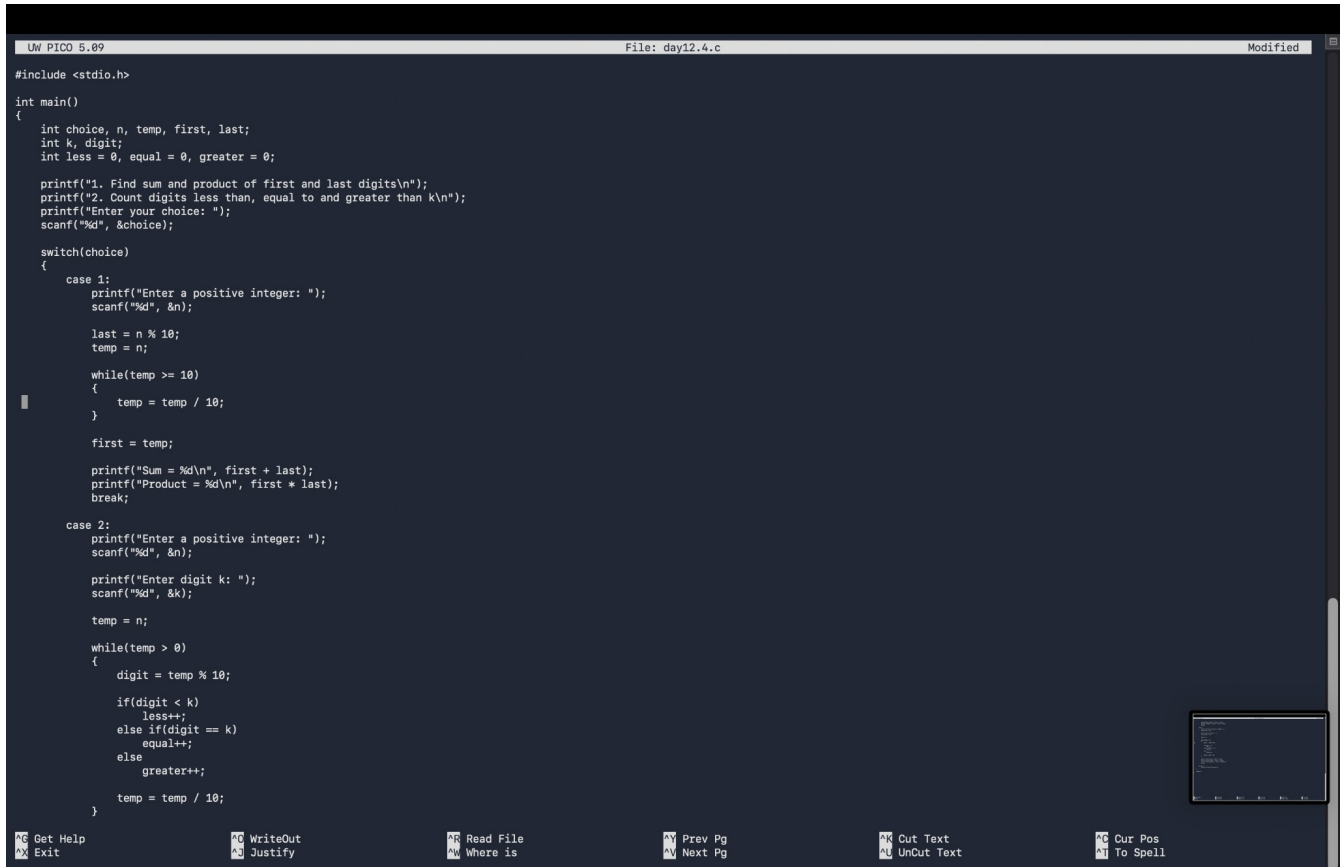
default:
    printf("Invalid choice\n");
}

return 0;
}
```

The editor has a dark theme and a status bar at the bottom with various keyboard shortcuts for editing and navigation.

Execution evidence for counting digits less than, equal to, and greater than k.

Complete Switch-Case Program - Part 1



```
UW PICO 5.09 File: day12.4.c Modified
#include <stdio.h>

int main()
{
    int choice, n, temp, first, last;
    int k, digit;
    int less = 0, equal = 0, greater = 0;

    printf("1. Find sum and product of first and last digits\n");
    printf("2. Count digits less than, equal to and greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            printf("Enter a positive integer: ");
            scanf("%d", &n);

            last = n % 10;
            temp = n;

            while(temp >= 10)
            {
                temp = temp / 10;
            }

            first = temp;

            printf("Sum = %d\n", first + last);
            printf("Product = %d\n", first * last);
            break;

        case 2:
            printf("Enter a positive integer: ");
            scanf("%d", &n);

            printf("Enter digit k: ");
            scanf("%d", &k);

            temp = n;

            while(temp > 0)
            {
                digit = temp % 10;

                if(digit < k)
                    less++;
                else if(digit == k)
                    equal++;
                else
                    greater++;

                temp = temp / 10;
            }
    }
}
```

Get Help Exit WriteOut Justify Read File Where is Prev Pg Next Pg Cut Text UnCut Text Cur Pos To Spell

The complete menu-driven program starts by displaying the two required choices and processes the selected case.

Complete Switch-Case Program - Part 2

```
OMYADAV@OMs-WINDOWS 11 ~ % clang day12.1.c -o day12.1
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1
Enter a positive integer: 1
Sum = 2
Product = 1
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1
Enter a positive integer: 3
Sum = 6
Product = 9
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1c
zsh: no such file or directory: ./day12 1c
OMYADAV@OMs-WINDOWS 11 ~ % ./day12.1
Enter a positive integer: 3
Sum = 6
Product = 9
OMYADAV@OMs-WINDOWS 11 ~ % █
```

Continuation of the switch-case program, including digit comparison and invalid-choice handling.

Conclusion

The HDOC Day-12 task was completed using a menu-driven C program with switch-case. The program performs both required operations and the test cases confirm the expected behavior.